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DIFFRACTION IN GRAVITATIONAL LENSING FOR COMPACT OBJECTS OF LOW MASS

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ABSTRACT

Diffraction effects are examined which are of possible importance in gravitational lensing by low-mass ($M \lesssim M_\odot$) objects. Instead of the usual approximation of geometrical optics to treat gravitational lensing, propagation of electromagnetic waves in the presence of a point mass is formulated in terms of a scalar wave equation which has the same form as the Schrödinger equation for Coulomb scattering of particles. Diffraction phenomena, such as the intensity of the image on the optical axis (which is infinite in the approximation of geometrical optics), the lowest mass for which significant brightening occurs, and a pattern of interference fringes, appear naturally in the well-studied solutions to this equation. Aspects of gravitational lensing involving diffraction are of possible interest in searches for dark matter in the form of low-mass stars and planets.

Subject headings: gravitation — interferometry

I. INTRODUCTION

The gravitational deflection of electromagnetic radiation and the imaging of distant, luminous sources has been extensively studied theoretically since the work of Liebes (1964) and Refsdal (1964) (for a recent review, see Peacock 1983). In recent years, several gravitational lenses apparently have been identified by the observation of multiple images of distant galaxies separated by arcseconds. A mass that is characteristic of a galaxy is then implied for the lens. It also has been noted that, although the separation of the images produced is not sufficient to be observed, even masses below 1 $M_\odot$ can produce significant lensing effects by intensifying the single image of the source. This property has been (e.g., Canizares 1982) and might be further utilized in searches for dark matter in the universe in the form of stellar or planetary objects with $M \lesssim 1\ M_\odot$. Such objects are one of the currently allowed forms for the dark matter (e.g., Carr, Bond, and Arnett 1984) which is inferred to be present from observations of stellar dynamics, galactic rotation, and the motion of galaxies in clusters.

Essentially all calculations have treated the propagation of the radiation in the approximation of geometrical optics—an excellent approximation for lensing by galaxies. For the $M \lesssim 1\ M_\odot$ class of lenses and radiation at radio wavelengths, however, it has been recognized that the effects of diffraction may not be completely negligible. Such effects might be valuable in the initial detection of low-mass objects or as the basis for alternative techniques in the verification of suspected lenses that might be detected by variations in the intensity of the image. The direct detection of low-mass objects by intensity variations alone does have the defect that the time scale for the variations tends to be long, and intensity fluctuations can be due to other causes. Diffraction phenomena that have been considered involve the degree of mutual coherence of the radiation (Mandzhos 1982; Schneider 1983) and the limiting low mass for intensification of the image (Bontz and Haugan 1981). Closely related to the latter is the intensity of the image produced by a point lens on the optical axis (Bliokh and Minakov 1985; Benson and Cooke 1979)—it is infinite in the approximation of geometrical optics.

The goal of the investigation here is to contribute to the understanding of diffraction effects in gravitational lensing by obtaining the "exact" solutions to the scalar wave equation that describes the propagation of electromagnetic radiation in the presence of a point mass. In § II we show that, in the weak-field limit which is appropriate here, the electromagnetic wave can be described by a scalar wave equation which has the same form as the Schrödinger equation for Coulomb scattering. This approach does not seem to have been taken previously. Considerable analytic information about the solutions of this wave equation is thus available which we utilize in § III to obtain an expression for the intensity of the image on the optical axis and to compare with solutions obtained in the approximation of geometrical optics. The necessary confluent hypergeometric functions are evaluated numerically in § IV to obtain the intensities, and thus to exhibit the diffraction patterns, for a range of relevant values for the parameters. Astronomical considerations—including especially the size of the source—are mentioned in § V.

II. SCALAR WAVE EQUATION

In the limit of a weak, static gravitational field due to a point mass M, the differential line element can be written to first order in isotropic coordinates for time t and space x^i as

$$ds^2 = g_{00}(cdt)^2 + \sum_i g_{ii}(dx^i)^2 , \tag{1}$$

where the metric $g_{\alpha\beta}$ at radius R is

$$g_{00} = -1 + \frac{2GM}{c^2R} , \tag{2a}$$

$$g_{ii} = 1 + \frac{2GM}{c^2R} \tag{2b}$$

for the spatial coordinates $i = 1, 2, 3$, and

$$g_{ij} = 0 \quad \text{for } i \neq j \tag{2c}$$

(e.g., Landau and Lifshitz 1962). The equations in general relativity analogous to Maxwell's equations are obtained from the

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electromagnetic field tensor $F^{\alpha\beta}$ and the relations

$$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0 \tag{3a}$$

and

$$F^{\alpha\beta};\beta = 0 \tag{3b}$$

for vacuum conditions. Here commas indicate ordinary derivatives, and semicolons denote covariant derivatives. For a static gravitational field, equations (3a) and (3b) are equivalent to the vector equations (Landau and Lifshitz 1962, p. 296)

$$\nabla \times \boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t}, \tag{4a}$$

$$\nabla \times \boldsymbol{H} = \frac{1}{c}\frac{\partial \boldsymbol{D}}{\partial t}, \tag{4b}$$

$$\nabla \cdot \boldsymbol{D} = 0, \tag{4c}$$

$$\nabla \cdot \boldsymbol{B} = 0, \tag{4d}$$

for the electromagnetic fields E, D, H, and B analogous to such fields in the absence of gravity. They are related by

$$\boldsymbol{D} = \boldsymbol{E}/\sqrt{-g_{00}} \tag{5a}$$

and

$$\boldsymbol{B} = \boldsymbol{H}/\sqrt{-g_{00}} \tag{5b}$$

The vector operations in equations (4a)–(4c) are performed in the three-dimensional space described by the metric g_{ij} $(i, j = 1, 2, 3)$.

By ignoring higher order terms in (GM/c^2R) and in (λ/R), where λ is the wavelength of the electromagnetic radiation, a scalar wave equation can then be obtained as a result of performing the usual operations,

$$\left(\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2}\right)\boldsymbol{E} = \frac{1}{c^2}\left(1 + \frac{4GM}{c^2R}\right)\frac{\partial^2 \boldsymbol{E}}{\partial t^2}. \tag{6}$$

As noted by others (e.g., Fock 1964), to lowest order the influence of gravity can be treated as an alteration in the index of refraction of the medium. The scalar wave equation can also be obtained from more general considerations (e.g., Fock 1964, p. 368). In equation (6), the spatial coordinates can be treated as ordinary Cartesian coordinates (e.g., Muhleman and Johnston 1966).

Substitution of the form

$$\boldsymbol{E}(R, t) = \psi(R)e^{i\omega t}\boldsymbol{e}, \tag{7a}$$

where $\boldsymbol{e}$ is a unit vector specifying the polarization and the wave number

$$k \equiv \omega/c, \tag{7b}$$

makes it possible to rearrange equation (6) into the familiar form of the time-independent Schrödinger equation that describes Coulomb scattering,

$$\nabla^2\psi + \left(k^2 - \frac{2nk}{R}\right)\psi = 0. \tag{8}$$

Here,

$$n \equiv -2GMk/c^2. \tag{9}$$

The problem of image formation by gravitational lenses can thus be reduced to solving equation (8) for astronomically suitable conditions.

We know that Coulomb scattering as described quantum mechanically by the analog of equation (8) goes to the correct classical limit for particles. Since this is also the correct form for the path of a ray in the approximation of geometrical optics, we are thus assured that equation (8) does in fact contain the proper limit for geometrical optics to the order in (GM/c^2R) that is ordinarily considered. This may be seen explicitly by obtaining the eikonal equation from equation (6) (e.g., Landau and Lifshitz 1962, p. 147) which coincides (to the specified order) with the well-known eikonal equation in the approximation of geometrical optics (Landau and Lifshitz 1962, p. 336). The usual solutions for the paths of the rays of radiation in the limit of geometrical optics then follow.

Because of this formal similarity in the deflection of electromagnetic waves by the gravitational lens and the quantum mechanical scattering of particles in the Coulomb potential, the extensive studies of equation (8) in the context of the latter problem can be brought to bear here. Of course, in the limit of geometrical optics the orbit of the light ray which is deflected by a point gravitational lens is the same as the Rutherford scattering of particles. From textbooks on quantum mechanics (e.g., Landau and Lifshitz 1977; Schiff 1968) the incident "plane waves" in equation (8) that describe radiation from a source that can be approximated as being at infinity are given by

$$\psi = \exp(-\pi n/2)\Gamma(1 + in)\exp(ikZ)F[-in, 1, ik(R - Z)], \tag{10}$$

where F is the confluent hypergeometric function and the wave propagates along the Z-axis. Here, the normalization is such that $\psi\psi^* = 1$ when $Z \to \infty$.

The usual limit that is taken in textbooks is $k(R-Z) \to \infty$, for which ψ approaches

$$\psi \to e^{i\boldsymbol{k}\cdot\boldsymbol{R} + in\ln kR(1-\cos\theta)} - \left[\frac{n\Gamma(1 + in)}{\Gamma(1 - in)}\frac{\exp[-in\ln(1/2)(1 - \cos\theta)]}{k(1 - \cos\theta)}\right] \times \frac{\exp(ikR - in\ln 2kR)}{R}. \tag{11}$$

The first term is the incident wave, and the second term is the scattered wave. The amplitude of the scattered wave gives the differential collisional cross section

$$\frac{d\sigma}{d\Omega} = \frac{n^2}{4k^2\sin^4(\theta/2)}, \tag{12}$$

which coincides with the classical Rutherford cross section. The limit to obtain equation (11) is not useful for gravitational lenses. It implies larger scattering angles at finite distances than are appropriate. For the gravitational lens, almost pure forward scattering is most important. We find no useful approximate forms for ψ when $k(R-Z) \nrightarrow \infty$.

III. WAVE SOLUTION FOR THE ELECTROMAGNETIC RADIATION SCATTERED BY A POINT MASS

The solution given by equation (11) describes the intensity of the electromagnetic radiation after it has been scattered by the gravitational lens. The incident wave is a plane wave with the phase modified by $in\ln kr(1 - \cos\theta)$, as usual, due to the long-range Coulomb or gravitational interaction. The source for the radiation is thus at an infinite distance, or at least much further

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than the observer, from the lens. When the source is located at a finite distance from the lens, as is usually more appropriate for actual astronomical situations, the curvature of the wave front of the incident wave must be taken into account. In exploring the basics of the wave effects in this paper, we limit our attention to the idealized case of an infinitely distant source. We expect to extend the solutions to the more general case in subsequent studies if warranted. When the wave length approaches zero, or $k \to \infty$, the plane wave solution of equations (10) or (11) should coincide with the solution which is derived classically in the ray approximation by Liebes (1964) and Refsdal (1964) for a point source of light at an infinite distance. At a glance the two waves expressed in equation (11) can be seen as two images produced by a gravitational lens with the R^{-1} potential of a point mass. However, the Liebes-Refsdal solution gives the intensities of the images as

$$I_1 = \tfrac{1}{4}(2 + \xi + \xi^{-1}) \tag{13a}$$

and

$$I_2 = \tfrac{1}{4}(-2 + \xi + \xi^{-1}) , \tag{13b}$$

where

$$\xi = \{1 + [\theta_0/(\theta/2)]^2\}^{1/2} \tag{14}$$

and

$$\theta_0{}^2 = \frac{4GM}{c^2 R_0} , \tag{15}$$

in which R_0 is the distance from the mass to the observer. Note that these expressions are obtained by taking the distance to the source to be infinite in the expressions of Liebes and Refsdal. Here, θ is the angle between the direction of propagation of the incident radiation and the line from the lens to the observer (see Fig. 1). It is the same θ as appears in equations (11) and (12). The scattering angles ζ_1 and ζ_2 defined by Figure 1 are

$$\zeta_1 = \zeta_2 = -\frac{1}{2}\theta + \left[\left(\frac{\theta}{2}\right)^2 + \theta_0{}^2\right]^{1/2} . \tag{16}$$

It is not easy to see that the solution given by equations (10) or (11) is coincident with the Liebes-Refsdal result in equations (13a), (13b), and (16) in the limiting case of $k \to \infty$. For example, the square of the amplitude of the incident wave in equation (11) stays constant at any angle, and that of the scattered wave diverges as θ^{-4} with decreasing θ. This behavior for the intensity of the radiation is not consistent with the angular dependence in equations (13a) and (13b), which diverges with θ^{-1}. The apparent inconsistency arises because the form of the solution in equation (11) is valid only for larger θ than those at which the gravitational lens effect is important. To see that the solutions are consistent, we must examine the original form given by equation (10).

In the following, we show how under certain restrictions the wave solution of equation (10) is consistent with the solution in the ray approximation. Even though we prove this only in a restricted sense, it is quite instructive in understanding the nature of the diffraction produced by a lens of low mass.

a) "Large" Angle Scattering

We start from a form for the wave function which gives the correct asymptotic expression at small angles. The confluent hypergeometric function can be written (e.g., Landau and Lifshitz 1977)

$$F(\alpha, \gamma, z) = \frac{\Gamma(\gamma)}{\Gamma(\gamma - \alpha)}(-z)^{-\alpha} G(\alpha, \alpha - \gamma + 1, -z) + \frac{\Gamma(\gamma)}{\Gamma(\alpha)} e^z z^{\alpha - \gamma} G(\gamma - \alpha, 1 - \alpha, z) , \tag{17}$$

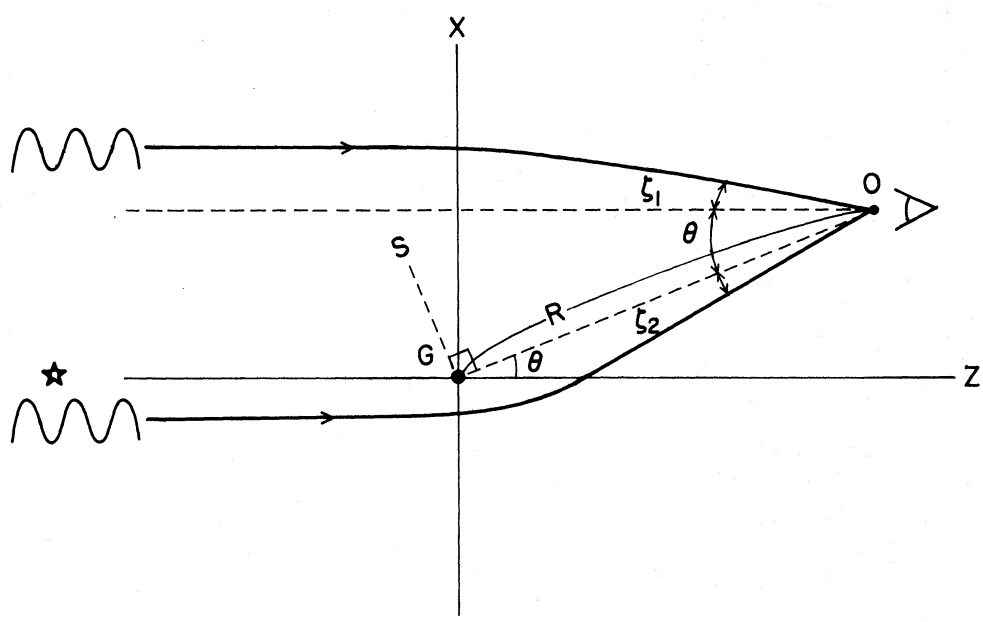

FIG. 1.—The gravitational lens G deflects the electromagnetic wave by angle ξ_1 and $\theta + \xi_2$. The observer is at point O at distance R from the lens.

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where function G is defined by

$$G(\alpha, \gamma, z) = \frac{\Gamma(1-\gamma)}{2\pi i} \int_{c_1} \left(1 + \frac{t}{z}\right)^{-\alpha} t^{\gamma-1} e^t dt \tag{18}$$

and c_1 is the path starting at $z = -\infty$, circling around the origin in a counterclockwise direction, and returning to the starting point. By expanding $(1 + t/z)^{-\alpha}$ in the integrand in powers of t/z, the asymptotic series

$$G(\alpha, \gamma, z) = 1 + \frac{\alpha\gamma}{1!\,z} + \frac{\alpha(\alpha+1)\gamma(\gamma+1)}{2!\,z^2} + \cdots + \frac{\alpha(\alpha+1)\cdots(\alpha+N-1)\gamma(\gamma+1)\cdots(\gamma+N-1)}{N!\,Z^N} + O(z^{-N-1}) \tag{19}$$

can be obtained. The series of equation (19) is not convergent, but nevertheless can be evaluated by taking a finite number of terms in the series (e.g., Morse and Feshbach 1953, p. 434).

The wave function (10) can be written in the form

$$\psi = e^{i\boldsymbol{k}\cdot\boldsymbol{R} + in \ln kR(1-\cos\theta)} G_1 - \left[\frac{n\Gamma(1+in)}{\Gamma(1-in)} \frac{e^{-in\ln(1/2)(1-\cos\theta)}}{k(1-\cos\theta)}\right] \frac{e^{ikR - in\ln 2kR}}{R} G_2 , \tag{20}$$

where

$$G_1 = G[-in, -in, -ik(R-Z)] \tag{21}$$

and

$$G_2 = G[1+in, 1+in, ik(R-Z)] . \tag{22}$$

Then, G_1 and G_2 represent the corrections to the amplitude and phase of the incident and scattered waves at small angles. When $\alpha\gamma/z \to 0$, G_1 and G_2 approach unity.

To see that the Liebes-Refsdal solution is consistent with equation (20), we construct a wave function from the solution obtained in the approximation of geometrical optics. We know the intensity of the radiation and the directions of propagation. We can thus write expressions for the waves that produce the images based on the Liebes-Refsdal solution and the WKB approximation. They are

$$\psi_I = \sqrt{I_1}\, e^{ip_1} \tag{23a}$$

and

$$\psi_S = \sqrt{I_2}\, e^{ip_2} \tag{23b}$$

The phases p_1 and p_2 can be determined from the directions of propagation

$$\left(\frac{\partial p_1}{\partial X}\right) \Big/ \left(\frac{\partial p_1}{\partial Z}\right) = -\tan\zeta_1 \tag{24a}$$

and

$$\left(\frac{\partial p_2}{\partial S}\right) \Big/ \left(\frac{\partial p_2}{\partial R}\right) = \tan\zeta_2 , \tag{24b}$$

where S is the coordinate perpendicular to the radius R (see Fig. 1). The origin for the phase can be taken arbitrarily here. The negative sign in equation (24a) appears because ζ_1 is taken as positive. Because of the approximation of small angles, $\tan\zeta \approx \zeta$, $\theta \gg \theta_0$, and

$$\frac{\partial p_1}{\partial Z} \approx \frac{\partial p_2}{\partial R} \approx k . \tag{25}$$

Then

$$p_1 = kZ + n\log t - \frac{n^2}{t} - \frac{n^3}{t^2} - \frac{5}{3}\frac{n^4}{t^3} + \cdots \tag{26a}$$

and

$$p_2 = kR - n\log t + \frac{n^2}{t} + \frac{1}{2}\frac{n^3}{t^2} + \frac{5}{3}\frac{n^4}{t^3} + \cdots , \tag{26b}$$

where

$$t = k(R-Z) . \tag{27}$$

If we substitute $\alpha = -in$ and $z = -ik(R-Z)$, the correction term for wave effects in the Liebes-Refsdal solution is

$$G_1^{\mathrm{LR}} = 1 + \frac{\alpha^2}{z} + \frac{\alpha^2(\alpha^2+2\alpha+1)}{2z^2} + \frac{\alpha^6+6\alpha^5+13\alpha^4+12\alpha^3}{6z^3} + O(z^{-4}) , \tag{28a}$$

whereas the wave solution equation (21) is

$$G_1 = 1 + \frac{\alpha^2}{z} + \frac{\alpha^2(\alpha^2+2\alpha+1)}{2z^2} + \frac{\alpha^6+6\alpha^5+13\alpha^4+12\alpha^3+\alpha^2}{6z^3} + O(z^{-4}) , \tag{28b}$$

(We notice the difference appears in the term $\alpha^2/6z^3$). Proceeding in a similar manner for the scattered wave gives

$$G_2^{\mathrm{LR}} = 1 - \frac{\alpha^2-2\alpha}{z} + \frac{\alpha^4-5\alpha^3+11\alpha^2}{2z^2} + O(z^{-3}) , \tag{29a}$$

whereas from equation (22)

$$G_2 = 1 - \frac{\alpha^2-2\alpha+1}{z} + \frac{\alpha^4-6\alpha^3+13\alpha^2-12\alpha+4}{2z^2} . \tag{29b}$$

The difference appears in the first-order term $1/z$.

The difference terms can be rewritten as

$$\frac{\alpha^2}{z^3} = \left(\frac{\theta_0}{\theta}\right)^4 \frac{1}{ik(R-Z)} , \tag{30a}$$

or

$$\frac{1}{z} = \frac{1}{ik(R-Z)} . \tag{30b}$$

The difference becomes significant when the typical scale length $R-Z$ is equal to the wavelength. This indicates that they are due to wave effects. Such differences are expected.

In summary, the Liebes-Refsdal solution agrees with the wave solution until the third order in z for incident wave and to first order in z for the scattered wave.

b) Small Angle Scattering

In the case of $\theta < \theta_0$ or $\alpha^2/z > 1$, the series in equation (19) cannot be used for the evaluation of G. It is then difficult to confirm that the wave solution is consistent with that of Liebes-Refsdal. We show here, instead, that the total intensity of the images given by the Liebes-Refsdal solution is consistent with that given by the solution of the wave equation. The confluent hypergeometric function can be expressed by the

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series

$$F(\alpha, \gamma, z) = 1 + \frac{\alpha z}{\gamma 1!} + \frac{\alpha(\alpha+1)z^2}{\gamma(\gamma+1)2!} + \cdots + \frac{\alpha(\alpha+1)\cdots(\alpha+N-1)z^N}{\gamma(\gamma+1)\cdots(\gamma+N-1)N!} + \cdots . \quad (31)$$

Using parameters

$$x \equiv \frac{z}{\alpha} = \left(\frac{\theta}{\theta_0}\right)^2 \quad (32a)$$

and

$$y \equiv -n = \frac{2GM}{c^2} k = \frac{\theta_0{}^2 kR}{2} , \quad (32b)$$

we have

$$F(-in, 1, z) = 1 + \frac{(-y^2x)}{(1!)^2} + \frac{(-y^2x)^2}{(2!)^2}\left(1 + \frac{1}{iy}\right) + \cdots + \frac{(-y^2x)^N}{(N!)^2}\left(1 + \frac{1}{iy}\right)\left(1 + \frac{2}{iy}\right)\cdots\left(1 + \frac{N-1}{iy}\right) + \cdots . \quad (33)$$

Of course, when $x \to 0$,

$$F(-in, 1, z) \to 1 . \quad (34)$$

This means that the intensity remains finite unlike in the Liebes-Refsdal solution.

To compare the solution from the wave equation with that from the approximation of geometrical optics, we take the limit of large k and hence large y. We limit our attention to $x \ll 1$, so that equation (33) can be evaluated by neglecting the terms $(N-1)/(iy)$:

$$F \to \sum_{N=0}^{\infty} \frac{(-y^2x)^N}{(N!)^2} = J_0(2y\sqrt{x}) , \quad (35)$$

where J_0 has the usual meaning for a Bessel function. This is valid because the main contributions to the series (33) come from $N \lesssim y\sqrt{x}$. Including the normalization term, we have

$$\psi\psi^* \approx \frac{2\pi y}{(1 - e^{-2\pi y})} J_0{}^2 , \quad (36)$$

where we have used the relation (Abramowitz and Stegun 1970)

$$|\Gamma(1 - iy)|^2 = \frac{\pi y}{\sinh(\pi y)} . \quad (37)$$

If we put

$$t \equiv y^2 x , \quad (38)$$

and neglecting $e^{-2\pi y}$, we get

$$\psi\psi^* \approx 2x^{-1/2}(\sqrt{\pi} t^{1/4} J_0)^2 . \quad (39)$$

The asymptotic form ($t \to \infty$) of the zeroth order Bessel function is

$$J_0(2\sqrt{t}) \to \frac{t^{-1/4}}{\sqrt{\pi}} \cos\left(2\sqrt{t} - \frac{\pi}{4}\right) , \quad (40)$$

so that the function $\pi^{1/2}t^{1/4}J_0[2(t)^{1/2}]$ approaches $\cos[2(t)^{1/2} - \pi/4]$. The function $\pi^{1/2}t^{1/4}J_0[2(t)^{1/2}]$ is shown in Figure 2. It rapidly approaches the asymptotic form.

It is easily seen that the total intensity $I_1 + I_2$ given by equations (13a) and (13b) is quite close to the value about which equation (39) oscillates. Both increase as

$$I_1 + I_2 \to x^{-1/2} \quad (41)$$

when $x \to 0$. The factor 2 in equation (39) appears because the average of the oscillating function $(\pi^{1/2}t^{1/4}J_0)^2$ is 1/2. The oscillation of the amplitude is, of course, due to the interference.

We should note that $\psi\psi^*$ has a maximum because $F \to 1$ when $x \to 0$. The maximum value is

$$(\psi\psi)^*_{\max} = \frac{2\pi y}{1 - e^{-2\pi y}} . \quad (42)$$

This is the upper limit for the amplification of the brightness of

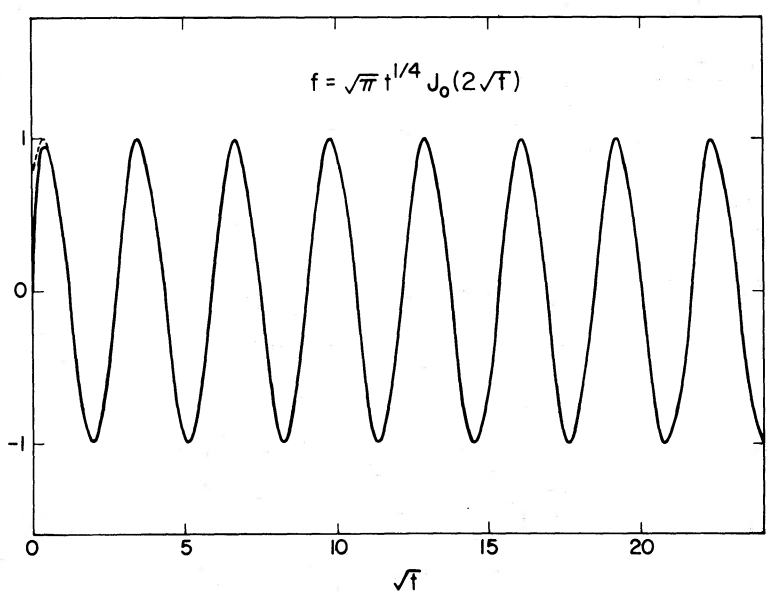

FIG. 2.—The function $\pi^{1/2}t^{1/4}J_0[2(t)^{1/2}]$. At large t, it can be approximated by $\cos[2(t)^{1/2} - \pi/4]$, which is continued to small t by the dashed line.

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the source by the gravitational lens. This approaches unity when $y \to 0$.

The maximum amplification has been obtained by Bliokh and Minakov (1975), and their expression (eq. [6] above) agrees with equation (42) when $y \gg 1$.

The fact that the luminosity of the lensed image approaches unity when $y \to 0$ means that in effect there is a minimum mass for the lens action. The condition that $y \gtrsim 1$ leads to

$$M \gtrsim 5 \times 10^{-7}\, M_{\odot}\, (\lambda/1 \text{ cm})\,. \tag{43}$$

For optical wave lengths, the minimum mass of the lens is $\sim 3 \times 10^{-11}\, M_{\odot}$. An approximate value for this critical mass has been obtained in a somewhat different context and with different methods by Bontz and Haugan (1981).

IV. NUMERICAL SOLUTION

The total luminosity for the point source can be calculated from equation (10). To obtain numerical values, we have solved the differential equation for the confluent hypergeometric function $F(-in, 1, z)$,

$$x \frac{d^2F}{dx^2} + (1 - iyx)\frac{dF}{dx} + y^2F = 0\,, \tag{44}$$

where x and y are used instead of n and z (eqs. [32a] and [32b]). The boundary conditions at $x = 0$ are

$$F = 1 \tag{45a}$$

and

$$\frac{dF}{dx} = -y^2\,. \tag{45b}$$

For the numerical calculations, we have rewritten equation (44) in the form

$$\frac{d}{dx}\begin{pmatrix} F \\ x\dfrac{dF}{dx} \end{pmatrix} = \begin{pmatrix} 0 & 1/x \\ -y^2 & iy \end{pmatrix}\begin{pmatrix} F \\ x\dfrac{dF}{dx} \end{pmatrix}. \tag{46}$$

The differential equation (46) can be solved by numerical integration.

The numerical value of F could also be calculated from the power series expansion (33). However, the term $(y^2x)^N/(N!)^2$ easily exceeds the number of available digits of the computer (e.g., 30 digits in the CDC cyber) when $y^2x \gg 1$, though the sum must remain near unity. Hence, we avoided this method.

The results are shown in Figure 3. When y is larger than one—that is, for larger masses or shorter wave lengths—the solution oscillates around the Liebes-Refsdal solution. The oscillations are, of course, the interference fringes. As y decreases, the oscillation becomes slower, and for $y = 0.3$ the fringe cannot be seen.

As shown in § III*b*, the luminosity of the image of a point source remains finite (eq. [42]). For the small y limit ($y \to 0$), the intensity approaches (see eq. [33])

$$(\psi\psi^*) = 1 - 2y^2x + \cdots\,. \tag{47}$$

When $x \ll \frac{1}{2}y^2$, the observed intensity of the source thus coincides with that which would occur without deflection by a lens. An indication of the overall behavior of the solution can be obtained from the matrix in equation (46)

$$\begin{pmatrix} 0 & 1/x \\ -y^2 & iy \end{pmatrix}. \tag{48}$$

The eigenvalues of this matrix are

$$\lambda = \frac{iy}{2}\left[1 \pm \left(1 + \frac{4}{x}\right)^{1/2}\right]. \tag{49}$$

When $x \ll 1$,

$$\lambda \approx \pm iy/x^{1/2}\,. \tag{50}$$

When $x \gg 1$,

$$\lambda_1 \approx iy \tag{51a}$$

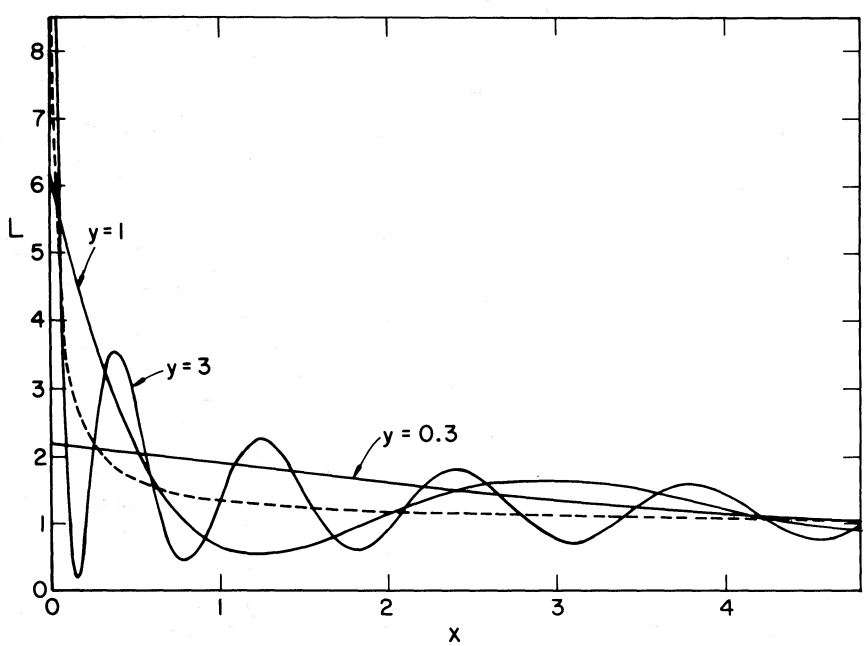

FIG. 3.—The intensification ($\psi\psi^*$) of the gravitationally lensed images (combined) as a function of x for various values of the parameter y. The dashed line shows the Liebes-Refsdal solution.

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and

$$\lambda_2 \approx -iy/x \,. \tag{51b}$$

Because the eigenvalues contain the variable x, we must regard the eigenvalues only as indicators of the variation of the solutions over distances shorter than that for appreciable change in the matrix of equation (48). Keeping this fact in mind, we can say that the solution oscillates as a function of $\lambda x \approx yx^{1/2}$ when $x \leq 1$. This variation is consistent with that found in the previous section, where the solution oscillates as a function of $t^{1/2} = yx^{1/2}$ (eq. [40]). When $x \gg 1$, equations (51a) and (51b) suggest that the intensity consists of the two variations with frequency λ_1 and λ_2.

The interference fringes produced by gravitational lensing are due to the phase difference between the rays propagating along different paths. The situation can easily be understood by analogy with the interference fringes produced by a double slit.

V. DISCUSSION OF ASTRONOMICAL CONSIDERATIONS

The interference treated in the foregoing sections is, of quite, quite analogous to that which is utilized by interferometers. The two rays that would be observed by two radio telescopes which are separated by some baseline distance in the absence of gravitational lensing are "conveniently" bent by the lens to arrive at a single location. The effective baseline is large—~ 1 AU for a lens due to 1 $M_\odot$ at a representative distance within our Galaxy.

As recognized by the previous investigators of coherence effects in gravitational lensing, its possible detection involves severe constraints on the astronomical conditions—mainly source size. Although our calculations are for the idealized case of a point source that is much more distant than the lens, approximate criteria can nevertheless be obtained in the context of our formulation. The large amplitude for the fringes is obtained only where $y \gtrsim 1$ and $x \lesssim 1$. We consider only this case. From either equation (40) or equation (50), the frequency for the oscillation of the intensity along the x-axis (i.e., as a function of distance perpendicular to the source-lens line) can be approximated as

$$\lambda_x \approx y/x^{1/2} \,, \tag{52}$$

for $x < 1$. When the source has a finite size and the telescope has a finite bandwidth, the condition for the fringes not to be appreciably altered is that the change in $(\lambda_x x)$ across the source and across the bandwidth satisfy

$$\Delta(\lambda_x x) \lesssim 2\pi \,. \tag{53}$$

Equation (53) can be transformed into approximate restrictions on source size,

$$\Delta\theta \lesssim 10^{-8}\left(\frac{\lambda}{10\text{ cm}}\right)\left(\frac{M_\odot}{M}\right)^{1/2}\left(\frac{1\text{ Mpc}}{R}\right)^{1/2} \text{arcsec} \,. \tag{54}$$

and on the bandwidth of the telescope

$$\frac{\Delta\lambda}{\lambda} \lesssim 3 \times 10^{-5}\left(\frac{\lambda}{10\text{ cm}}\right)\left(\frac{M_\odot}{M}\right)\left(\frac{\theta_0}{\theta}\right)^{1/2} \,, \tag{55}$$

where λ is the wavelength of the radiation that is being detected. The observed brightness of the lensed source will vary because the observer passes through the diffraction pattern as a result of the transverse relative velocity V of Earth, the lens, and the source. The time scale for this variation is

$$T \approx \frac{2\pi}{\lambda_x \dot{x}} = 100\left(\frac{300\text{ km s}^{-1}}{V}\right)\left(\frac{M_\odot}{M}\right)^{1/2} \times \left(\frac{R}{1\text{ Mpc}}\right)^{1/2}\left(\frac{\lambda}{1\text{ cm}}\right) \text{s} \,. \tag{56}$$

The most severe limitation tends to follow from equation (54). For example, if a cosmological distance and a 10^{-4} $M_\odot$ lens is considered, then the source size must be $\sim 10^{15}$ cm for observations at $\lambda = 10$ cm. This is, perhaps, appropriate for the cores of quasars, as indicated by the variations in the optical light, but seems to be considerably smaller than is likely for the size of the radio emitting region. If, however, the lens is nearby ($R \lesssim 1$ Mpc), then equation (54) can reasonably be satisfied for distant quasars with radio sizes of $\sim 10^{17}$ cm and masses for the lens $M \gtrsim 10^{-4}$ $M_\odot$. Another possible source is the water masers in other galaxies, such as those that have recently been detected (Claussen, Heiligman, and Lo 1984; Henkel *et al.* 1984). If they are composed of maser spots comparable in size ($\sim 10^{13}$ cm) to those in our own Galaxy (Walker, Matsakis, and Garcia-Barreto 1982; Deguchi 1982) and sources are found at, for example, 100 Mpc, then equation (54) could be satisfied for $M \lesssim 10^{-2}$ $M_\odot$ and a lens at ~ 1 Mpc. The time scale T in equation (56) would then be ~ 10 minutes. Astrophysical masers also have small bandwidths $\Delta\lambda/\lambda$. Equation (54) does represent a strong limit. Smaller interference effects and hence larger $\Delta\theta$ may well be detectable by examining the autocorrelation function, as discussed previously (Mandzhos 1981; Schneider 1983). We emphasize that the restrictions of equations (54) and (55) are only approximate and subject to the requirement that $x \ll 1$ is already satisfied. We intend to deal with these issues in more detail in a subsequent contribution.

Another issue is, of course, the probability of alignment of Earth with the lens and source. For a lensing mass M to be within an angle θ_0 of the line of sight to a source at a distance L, the probability is

$$P \approx \left(\frac{\rho}{M}\right)\theta_0{}^2 L^3 \approx \frac{4G\rho L^2}{c^2} \,, \tag{57}$$

where ρ is the average mass density of objects of mass M along the line of sight. Unless L is a cosmological distance and ρ is near the critical mass density of 10^{-29} g cm^{-3}, the probability given by equation (57) is quite small for other scenarios of likely astronomical interest. The low probability can be at least partly compensated for by utilizing a large number of sources, such as the thousands of spots of which nonstellar water masers seem to be composed (at optical wavelengths for which diffraction effects are unimportant, large numbers of nearby stars might be monitored). The result of equation (57) can also be improved by noting that, in considering low mass ($M \leq 10^{-2}$ $M_\odot$) lenses within our Galaxy, the rotation of the Galaxy causes the line of sight to an extragalactic source to change by θ_0 in a time scale of days.

In summary, the chief results of our study are solutions for the propagation of electromagnetic radiation under the influence of a point mass that retain the wave character of the radiation. Diffraction effects thus appear which are ignored in the usual, geometrical optics approximation. Such effects are of potential importance in gravitational lensing by low (i.e., stellar) masses. An analytic expression for the intensification

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when the source, lens, and observer are aligned—which is formally infinite in the approximation of geometrical optics—arises naturally. Only the short-wavelength or high mass limit of this expression seems to have been obtained previously (Bliokh and Minakov 1975). A minimum mass for appreciable intensification also follows from this analytic expression, an estimate for which was obtained previously by Bontz and Haugen (1981). A formulation of the lensing problem in terms of solutions to a scalar wave equation which has the well-known form of (nonrelativistic) quantum mechanical, Coulomb scattering does not seem to have been presented previously. Diffraction effects do, of course, appear automatically in this approach.

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REFERENCES

Abramowitz, M., and Stegun, I. A. 1970, *Handbook of Mathematical Functions* (New York: Dover).
Benson, J. R., and Cooke, J. H. 1979, *Ap. J.*, **227**, 360.
Bliokh, P. V., and Minakov, A. A. 1975, *Ap. Space Sci.*, **34**, L7.
Bontz, R. J., and Haugan, M. P. 1981, *Ap. Space Sci.*, **78**, 199.
Canizares, C. R. 1982, *Ap. J.*, **263**, 508.
Carr, B. J., Bond, J. R., and Arnett, W. D. 1984, *Ap. J.*, **277**, 445.
Claussen, M. J., Heiligman, G. M., and Lo, K. Y. 1984, *Nature*, **310**, 298.
Deguchi, S. 1982, *Ap. J.*, **259**, 634.
Fock, V. 1964, *The Theory of Space, Time, and Gravitation* (New York: Macmillan).
Henkel, C., Güsten, R., Downes, D., Thum, C., Wilson, T. L. and Biermann, P. 1984, *Astr. Ap.*, **141**, L1.
Landau, L. D., and Lifshitz, E. M. 1962, *Classical Theory of Fields* (Oxford: Pergamon).
Landau, L. D., and Lifshitz, E. M. 1977, *Quantum Mechanics* (3d ed.; Oxford: Pergamon), p. 560.
Liebes, S. 1964, *Phys. Rev.*, **B133**, 835.
Mandzhos, A. V. 1981, *Soviet Astr. Letters*, **7**, 213.
Morse, P. M., and Feshbach, H. 1953, *Methods of Theoretical Physics* (New York: McGraw-Hill).
Muhleman, D. O., and Johnston, I. D. 1966, *Phys. Rev. Letters*, **17**, 455.
Peacock, J. A. 1983, in *Quasars and Gravitational Lenses: Proc. the 24th Liège Internat. Ap. Colloquium* (Liège: Université de Liège), p. 98.
Refsdal, S. 1964, *M.N.R.A.S.*, **128**, 295.
Schiff, L. I. 1968, *Quantum Mechanics* (3d ed.; New York: McGraw-Hill), p. 138.
Schneider, P. 1983, in *Quasars and Gravitational Lenses: Proc. 24th Liège Internat. Ap. Colloquium* (Liège: Université de Liège), p. 131.
Walker, R. C., Matsakis, D. N., and Garcia-Barreto, J. A. 1982, *Ap. J.*, **255**, 128.

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